Supplementary Information: Perovskite Solar Cell Performance Prediction with LLM-Derived Feature Augmentation under Leakage-Aware DOI Evaluation

Table S1. Predictors per tier

Column sets shown for a preprocessor fitted on the full filtered data (documentation); in the experiments the `_na` indicators and the >80% missing filter are re-derived from each training partition. Tiers are nested: $t_1 \subset t_2 \subset t_3$.

t1: 77 predictors (77 added relative to the previous tier)

process (21): ETL_deposition_method, HTL_deposition_method, Perovskite_deposition_method, Perovskite_thickness_nm, ETL_thickness_nm, HTL_thickness_nm, Backcontact_thickness_nm, Perov_anneal_temp_C, Perov_anneal_time_min, Perov_deposition_ Perovskite_dimension_list_of_layers, Perov_quenching_crystallisation, Perovskite_dimension_3D_with_2D_capping_layer, Perovskite_deposition_solvent_annealing, Perovskite_thickness_nm_na, ETL_thickness_nm_na, HTL_thickness_nm_na, Backcontact_thickness_nm_na, Perov_anneal_temp_C_na, Perov_anneal_time_min_na, Perovskite_dimension_list_of_layers_na

device (9): Cell_architecture, Perovskite_band_gap_measured, is_flexible_x, is_semitransparent, Encapsulation, Add_lay_front, Perovskite_pl_max_nm, Perovskite_band_gap_measured_na, Perovskite_pl_max_nm_na

composition (47): num_a_ions, num_b_ions, num_c_ions, total_ion_count, a_coeff_sum, b_coeff_sum, c_coeff_sum, a_FA_fraction, a_MA_fraction, a_Cs_fraction, a_Rb_fraction, a_organic_fraction, a_other_fraction, b_Pb_fraction, b_Sn_fraction, b_Bi_fraction, b_other_fraction, halide_I_fraction, halide_Br_fraction, halide_Cl_fraction, halide_F_fraction halide_other_fraction, is_inorganic, is_leadfree, is_3D, is_2D, is_2D3D_mixture, Perovskite_composition_perovskite_inspired_ Perovskite_composition_perovskite_ABC3_structure, a_coeff_sum_na, b_coeff_sum_na, c_coeff_sum_na, a_FA_fraction_na, a_MA_fraction_na, a_Cs_fraction_na, a_Rb_fraction_na, a_organic_fraction_na, a_other_fraction_na, b_Pb_fraction_na, b_Sn_fraction_na, b_Bi_fraction_na, b_other_fraction_na, halide_I_fraction_na, halide_Br_fraction_na, halide_Cl_fraction_na halide_F_fraction_na, halide_other_fraction_na

t2: 179 predictors (102 added relative to the previous tier)

material (102): ETL_band_gap, ETL_LUMO, ETL_HOMO, ETL_electron_mobility, ETL_hole_mobility, ETL_dielectric_constant, ETL_min_atomic_mass, ETL_max_atomic_mass, ETL_avg_atomic_mass, ETL_min_radius, ETL_max_radius, ETL_avg_radius, ETL_en_min, ETL_en_max, ETL_en_avg, HTL_band_gap, HTL_LUMO, HTL_HOMO, HTL_electron_mobility, HTL_hole_mobility, HTL_dielectric_constant, HTL_min_atomic_mass, HTL_max_atomic_mass, HTL_avg_atomic_mass, HTL_min_radius, HTL_max_radius, HTL_avg_radius, HTL_en_min, HTL_en_max, HTL_en_avg, Perovskite_band_gap, Perovskite_LUMO, Perovskite_HOMO, Perovskite_electro Perovskite_hole_mobility, Perovskite_dielectric_constant, Perovskite_min_atomic_mass, Perovskite_max_atomic_mass, Perovskite_avg_atomic_mass, Perovskite_min_radius, Perovskite_max_radius, Perovskite_avg_radius, Perovskite_en_min, Perovskite_en_max, Perovskite_en_avg, perovskite_tolerance_factor, perovskite_octahedral_factor, perovskite_cation_radius_ perovskite_tau_descriptor, HTL_hole_barrier, ETL_electron_barrier, ETL_band_gap_na, ETL_LUMO_na, ETL_HOMO_na, ETL_electron_mobility_na, ETL_hole_mobility_na, ETL_dielectric_constant_na, ETL_min_atomic_mass_na, ETL_max_atomic_mass_na ETL_avg_atomic_mass_na, ETL_min_radius_na, ETL_max_radius_na, ETL_avg_radius_na, ETL_en_min_na, ETL_en_max_na, ETL_en_avg_na, HTL_band_gap_na, HTL_LUMO_na, HTL_HOMO_na, HTL_electron_mobility_na, HTL_hole_mobility_na, HTL_dielectric_con HTL_min_atomic_mass_na, HTL_max_atomic_mass_na, HTL_avg_atomic_mass_na, HTL_min_radius_na, HTL_max_radius_na, HTL_avg_radius_na, HTL_en_min_na, HTL_en_max_na, HTL_en_avg_na, Perovskite_band_gap_na, Perovskite_LUMO_na, Perovskite_HOMO_ Perovskite_electron_mobility_na, Perovskite_hole_mobility_na, Perovskite_dielectric_constant_na, Perovskite_min_atomic_ma Perovskite_max_atomic_mass_na, Perovskite_avg_atomic_mass_na, Perovskite_min_radius_na, Perovskite_max_radius_na, Perovskite_avg_radius_na, Perovskite_en_min_na, Perovskite_en_max_na, Perovskite_en_avg_na, perovskite_tolerance_factor_na perovskite_octahedral_factor_na, perovskite_cation_radius_ratio_na, perovskite_tau_descriptor_na, HTL_hole_barrier_na, ETL_electron_barrier_na

t3: 223 predictors (44 added relative to the previous tier)

literature (44): has_BCP, has_LiF, has_PEAI, has_SAM, has_C60, has_PCBM, has_Spiro, has_PEDOT, has_NiOx, has_SnO2, has_TiO2, has_ZnO, has_PTAA, has_fullerene, has_passivation_layer, is_flexible_y, is_encapsulated, has_stability_measurement, used_antisolvent, used_glovebox, used_vacuum_evaporation, json_numeric_values_found, has_BCP_na, has_LiF_na, has_PEAI_na, has_SAM_na, has_C60_na, has_PCBM_na, has_Spiro_na, has_PEDOT_na, has_NiOx_na, has_SnO2_na, has_TiO2_na, has_ZnO_na, has_PTAA_na, has_fullerene_na, has_passivation_layer_na, is_flexible_y_na, is_encapsulated_na, has_stability_measurement_na used_antisolvent_na, used_glovebox_na, used_vacuum_evaporation_na, json_numeric_values_found_na

Table S2. Per-split results of the matched CatBoost ablation (DOI-grouped)

Subset	Tier	Split	n_{test}	R^2	MAE	RMSE
nip	t1	1	5476	0.3524	3.101	3.937
nip	t1	2	5376	0.3780	3.067	3.928
nip	t1	3	5688	0.3845	3.068	3.936
nip	t1	4	5785	0.3923	2.983	3.814
nip	t1	5	5736	0.3652	3.091	3.976
nip	t2	1	5476	0.4006	2.931	3.788
nip	t2	2	5376	0.4228	2.933	3.784
nip	t2	3	5688	0.4030	2.999	3.876
nip	t2	4	5785	0.4049	2.957	3.775
nip	t2	5	5736	0.3877	3.007	3.905
nip	t3	1	5476	0.3939	2.960	3.809
nip	t3	2	5376	0.4197	2.953	3.794
nip	t3	3	5688	0.4091	2.982	3.856
nip	t3	4	5785	0.4071	2.943	3.768
nip	t3	5	5736	0.3963	2.981	3.877
overall	t1	1	8224	0.3707	3.082	3.914
overall	t1	2	8505	0.3345	3.101	3.958
overall	t1	3	8269	0.3604	3.136	3.945
overall	t1	4	8550	0.3990	3.036	3.859
overall	t1	5	8594	0.3767	3.051	3.862
overall	t2	1	8224	0.4145	2.943	3.775
overall	t2	2	8505	0.3712	2.985	3.847
overall	t2	3	8269	0.3787	3.067	3.888
overall	t2	4	8550	0.4420	2.892	3.719
overall	t2	5	8594	0.4326	2.864	3.684
overall	t3	1	8224	0.4089	2.966	3.793
overall	t3	2	8505	0.3731	2.982	3.842
overall	t3	3	8269	0.3782	3.075	3.890
overall	t3	4	8550	0.4408	2.893	3.723
overall	t3	5	8594	0.4317	2.874	3.687
pin	t1	1	2437	0.3046	3.116	3.919
pin	t1	2	2521	0.1653	3.166	4.151
pin	t1	3	2723	0.2863	3.123	4.020
pin	t1	4	2872	0.2065	3.202	4.200
pin	t1	5	2441	0.2596	3.231	4.134
pin	t2	1	2437	0.3447	3.026	3.804
pin	t2	2	2521	0.2043	3.081	4.053
pin	t2	3	2723	0.3437	2.994	3.855
pin	t2	4	2872	0.2379	3.105	4.116
pin	t2	5	2441	0.3248	3.028	3.948
pin	t3	1	2437	0.3592	2.980	3.762
pin	t3	2	2521	0.1815	3.136	4.111
pin	t3	3	2723	0.3385	3.009	3.871
pin	t3	4	2872	0.2571	3.105	4.064
pin	t3	5	2441	0.3315	3.026	3.928

Table S3. Split integrity (all 270 recorded fits)

DOI-grouped fits: 135; maximum DOI overlap between test and train: 0, validation and train: 0, test and validation: 0. Records without DOI: 232 (one block). Publications (normalised DOIs): 7326.

Data SHA-256: `600efce587a1bd81c2a192e36db6534d8d78a3edfadb635dd40e685c9314ffdc`; config SHA-256: `dddb51a44e89fde8`; git HEAD at run time: `9b1d2ca3e1c7` (working tree dirty: True); CatBoost 1.2.10, scikit-learn 1.6.1, Python 3.13.5.

Table S4. Literature-feature coverage and constancy within publication

Feature	coverage	constant within DOI
contact_thickness_nm	0.176	1.000
etl_anneal_temp_C	0.053	1.000
etl_thickness_nm	0.101	1.000
has_BCP	0.362	1.000
has_C60	0.362	1.000
has_LiF	0.362	1.000
has_NiOx	0.362	1.000
has_PCBM	0.362	1.000
has_PEAI	0.362	1.000
has_PEDOT	0.362	1.000
has_PTAA	0.362	1.000
has_SAM	0.362	1.000
has_SnO2	0.362	1.000
has_Spiro	0.362	1.000
has_TiO2	0.362	1.000
has_ZnO	0.362	1.000
has_fullerene	0.362	1.000
has_passivation_layer	0.362	1.000
has_stability_measurement	0.362	1.000
htl_anneal_temp_C	0.037	1.000
htl_thickness_nm	0.081	1.000
is_encapsulated	0.362	1.000
is_flexible_y	0.362	1.000
json_numeric_values_found	0.362	1.000
perov_anneal_temp_C	0.070	1.000
perov_anneal_time_min	0.063	1.000
perov_concentration	0.051	1.000
perovskite_thickness_nm	0.122	1.000
spin_speed_rpm	0.100	1.000
used_antisolvent	0.362	1.000
used_glovebox	0.362	1.000
used_vacuum_evaporation	0.362	1.000

Coverage values below the median belong to sparsely reported process parameters; all other columns are non-missing exactly for the 14,896 records of publications with an extraction record.

Extraction system prompt (verbatim from the recovered script)

```
You are a materials-science data-extraction assistant.

Given a Markdown document converted from a published paper, extract every
material reported with its measured or computed properties.

Return ONLY a single JSON object (no markdown fences, no commentary) with this
top-level schema:

{
  "source": "DOI: <doi-from-paper>",
  "materials": [
    {
```

```
      "formula": "<chemical formula>",
      "common_name": "<human-readable name>",
      "type": "<e.g. perovskite, ETL, HTL, metal contact>",
      ... <all properties you can find: band_gap, PV_performance, deposition_method, etc.>
    }
  ]
}

Rules:
- Always include units and measurement method when available.
- Include error bars / standard deviations if reported.
- For PV performance, separate champion vs average and HTM-free vs with-HTM.
- If the DOI is embedded in the document, use it for the "source" field.
- Do NOT invent data. Only extract what is explicitly stated.
- Return raw JSON only - no surrounding text, no code fences.
```

Model-development grid

CatBoost depth/L2 pairs: (6,3), (8,3), (10,3), (8,10), (10,10), (8,1); learning rate 0.05; up to 3,000 iterations; early stopping 150 rounds on the validation set; three seeds (42, 7, 123) for the selected configuration; XGBoost: depth 8, learning rate 0.05, 1,500 trees, subsample 0.8, colsample 0.7, $\lambda = 3$, early stopping 100 rounds.